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Brand Equity, Brand Preference, and Purchase Intent

Cathy J. Cobb-Walgren, Cynthia A. Ruble and Naveen Donthu

The issue of brand equity has emerged as one of the most critical areas for marketing management in the 1990s. Despite strong interest in the subject, however, there is little empirical evidence of how brand value is created and what its precise effects are. This study explores some of the consequences of brand equity. In particular, the authors examine the effect of brand equity on consumer preferences and purchase intentions. For comparative purposes, two sets of brands are tested, one from a service category characterized by fairly high financial and functional risk (hotels), and one from a generally lower risk product category (household cleansers). Each set includes two brands that are objectively similar (based on Consumer Reports ratings), but they have invested markedly different levels of advertising spending over the past decade. Across both categories, the brand with the higher advertising budget yielded substantially higher levels of brand equity. In turn, the brand with the higher equity in each category generated significantly greater preferences and purchase intentions.

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Introduction

Factories rust away,
packages become obsolete,
products lose their relevance.
But great brands live forever.

—Backer Spielvogel Bates

The cost of bringing a new brand to market is approximately \$100 million (Ourusoff 1992), with a 50 percent probability of failure (Crawford 1993). Thus, it is not surprising that companies seeking growth opportunities may prefer to acquire existing brands. Consider the large number of corporate mergers and leveraged buyouts that have occurred in recent years. In 1985, Philip Morris paid \$5.6 billion for General Foods. Three years later, the cigarette giant acquired the assets of Kraft for nearly \$13 billion—six times book value (Morgenson 1991). Grand Metropolitan reportedly paid almost \$6 billion for Pillsbury in 1989 (Marcom 1989). In each case, the purchase price reflected far more than the factories or the physical product produced therein. These corporate mergers were also about the acquisition of intangible assets—namely, brands.

To the consumer on Main Street, the terms “product” and “brand” are often used interchangeably. But Wall Street and Madison Avenue know the difference. A product is “something that offers a functional benefit” (Farquhar 1989, p. 24). A brand, on the other hand, is “a name, symbol, design, or mark that enhances the value of a product beyond its functional value” (Farquhar 1989, p. 24). Consider the case of the Quaker Oats brand. In 1991, the retail price of Quaker Oats oatmeal was 3,000 percent higher than the price of its basic ingredient, despite the fact that oats are essentially a commodity product, the wholesale price of which decreased 33 percent between 1980 and 1990 (Morgenson 1991).

Why are businesses and consumers alike willing to pay so much for brand names? Stated simply, brand names add value. The added value that a

brand name gives to a product is now commonly referred to as "brand equity" (Aaker 1991).

Brand equity can be discussed from the perspective of the investor, the manufacturer, the retailer, or the consumer. Clearly, brand names add value to each of these groups. Investors have a financial motivation for extracting the value of a brand name from the value of a firm's other assets. Manufacturers and retailers, on the other hand, are motivated more by the strategic implications of brand equity (Keller 1993). To the manufacturer, brand equity affords a differential advantage that enables the firm to generate greater volume and greater margins. Brand equity provides a strong platform for introducing new products and insulates the brand against competitive attacks. From the perspective of the trade, brand equity contributes to the overall image of the retail outlet. It builds store traffic, ensures consistent volume, and reduces risk in allocating shelf space.

None of this is meaningful, however, if the brand has no meaning to the consumer. In other words, there is value to the investor, the manufacturer, and the retailer *only if* there is value to the consumer (Farquhar 1989; Crimmins 1992). Thus, it is important to understand how brand value is created in the mind of the consumer and how it translates into choice behavior.

The purpose of the present study is twofold: 1) to measure the equity of brands which vary along selected criteria; and 2) to investigate the impact of brand equity on brand preferences and purchase intentions. The study examines the equity of both products and services, since the existing work on brand equity has focused almost exclusively on products and has failed to adequately consider service industries (Smith 1991). In addition, we include categories which differ in functional and financial risk, since these factors may influence brand equity (Landor Associates 1990). Most importantly, we include brands which are highly similar on physical attributes (as rated by *Consumer Reports*), but which vary significantly in the level of advertising support. Our reasoning is that advertising is the primary mechanism for creating psychological differentiation among brands and for enhancing brand equity (Ryan 1991; Aaker and Biel 1993). The specific brands under investigation include Holiday Inn and Howard Johnson for the hotel category and Soft Scrub and Bon Ami for the household cleanser category.

Literature Review

Since the term "brand equity" emerged in the 1980s, there has been a burgeoning interest in the subject among marketing academicians and practitioners. As evidence, a 1991 survey of Marketing Science Institute members ranked brand equity the number one issue facing marketing management (Aaker 1991). Researchers have focused primarily on defining and measuring the concept and, to a lesser extent, understanding its causes and effects. This literature review will address both the measurement and management of brand equity.

Measurement of Brand Equity

Much of the writing on brand equity has been concerned with definitional issues. As Crimmins (1992) observed, accurate measurement of brand equity must precede effective equity management. It is difficult to manage "added value" without knowing the actual value that a brand name adds to a product.

A number of alternative methods have been suggested for measuring brand equity. The techniques tend to be either financial or consumer-related. Among the financial measures, Simon and Sullivan (1993) used movements in stock prices to capture the dynamic nature of brand equity, on the theory that the stock market reflects future prospects for brands by adjusting the price of firms.

Mahajan, Rao and Srivastava (1991) used the potential value of brands to an acquiring firm as an indicator of brand equity. Another financial measure (applicable only when launching a new product) is based on brand replacement, or the requirements for funds to establish a new brand, coupled with the probability of success (see Simon and Sullivan 1993).

One of the most publicized financial methods is used by *Financial World* in its annual listing of worldwide brand valuation (Ourusoff 1993). FW's formula calculates net brand-related profits, then assigns a multiple based on brand strength (defined as a combination of leadership, stability, trading environment, internationality, ongoing direction, communication support, and legal protection).

Within the marketing literature, operationalizations of brand equity usually fall into two groups: those involving consumer perceptions (e.g., awareness, brand associations, perceived quality) and those involving consumer behavior (e.g., brand loyalty, will-

ingness to pay a high price). Among the perceptual measures, one technique uses consumer preference ratings for a branded product versus an unbranded equivalent (see Aaker 1991). Another approach, used by several authors, treats brand equity as brand name importance, since the name of a brand is often its core indicator (Louviere and Johnson 1988; Yovovich 1988; Sharkey 1989; MacLachlan and Mulhern 1991).

Keller (1993, p. 2) offered the following perceptual definition: "the differential effect of brand knowledge on consumer response to the marketing of the brand." Brand knowledge was decomposed into brand awareness (recall and recognition) and brand image (a combination of the favorability, strength, and uniqueness of brand associations).

Among the methods relying more on consumer behavior, Kamakura and Russell (1993) used scanner data to come up with three measures of brand equity. Their first measure—perceived value—was defined as the value of the brand which cannot be explained by price and promotion. Their second measure—brand dominance ratio—provided an objective value of the brand's ability to compete on price. Their third measure—intangible value—was operationalized as the utility perceived for the brand minus objective utility measurements.

Aaker (1991) is one of the few authors to incorporate both attitudinal and behavioral dimensions in his definition. He suggested using a brand-earnings multiplier that is based on a weighted average of the brand on five key components of brand equity (awareness, associations, perceived quality, loyalty, and other proprietary assets such as patents and trademarks).

There are some advantages to combining both consumer perceptions and actions into a single marketing measure of brand equity. It is well documented that attitudes alone are generally a poor predictor of marketplace behavior. On the other hand, consumer perceptions are clearly a precursor to behavioral manifestations of brand equity. And, as Biel (1992, pp. RC7-RC8) observed, "[C]onsumer behavior is, at root, driven by perceptions of a brand. While behavioral measures of purchase describe the existence of equity, they fail to reveal what is in the hearts and minds of consumers that is actually driving equity."

Despite the large number of alternatives suggested in the literature, no single measure is ideal. According to Lipman (1989), there is not even agreement on the relative strengths and weaknesses of each. Financial theorists, such as Simon and Sullivan, argue that the best techniques for measuring brand equity rely on objective, market-based data which allow com-

parisons over time and across firms. To them, the use of consumer attitudes and preferences is flawed due to their inherent subjectivity. Many marketers argue, on the other hand, that while brands do have value to various constituencies, it is the *consumer* who first determines brand equity (Farquhar 1989; Crimmins 1992). The present study relies on a consumer-based, perceptual measure of brand equity.

Antecedents and Consequences of Brand Equity

How exactly is brand value created? There is broad-based agreement that one of the major contributors to brand equity is advertising (Aaker and Biel 1993). According to Prentice (as cited in Ryan 1991, p. 19):

The consumer's perception of brand value comes from many sources, but essentially it is based on ideas—rational or emotional—that set the brand apart from competitive brands. What kinds of marketing activities implant these ideas about a brand's uniqueness in the mind?....Advertising is the most common.

Advertising can influence brand equity a number of ways. It can create awareness of the brand and increase the probability that the brand is included in the consumer's evoked set. It can contribute to brand associations which, when stored in accessible memory, translate into "nonconscious but reliable behavioral predispositions" (Krishnan and Chakravarti 1993, p. 214). Advertising can affect the perceived quality of a brand, and it can influence usage experience. No published studies to date have examined all of the components of brand equity as they relate to advertising. But a few studies have looked at isolated aspects of brand equity.

Stigler (1961) found that advertising which provides information about objective attributes such as price and physical traits will influence brand associations. Light (1990) reported a correlation between advertising spending and perceived quality, but not between promotional weight and the perception of quality. Nelson (1974) demonstrated that heavy advertising can improve perceived quality for experience goods, which by definition are difficult to evaluate prior to purchase. Most of the information content of such advertising, Nelson suggested, is carried in the brand name. In a series of experiments, Kirmani and Wright (1989) found that the perceived expense of a brand's advertising campaign can influence consumers' expectations of product quality.

Advertising can make positive brand evaluations

and attitudes readily accessible in memory (Farquhar 1989). This is crucial to the development of brand equity because, as Herr and Fazio (1992) noted, favorable brand attitudes will only guide perceptions and behavior if those attitudes can be instantly evoked.

Advertising also influences behavioral manifestations of brand equity. There is a long history of research on the relationship between advertising and sales. The ongoing PIMS project has confirmed a correlation between share of market and share of voice. On average, market leaders spend 20 percent more of their budgets on advertising than do their nearest competitors (Kim 1990). Prentice (as cited in Ryan 1991) compared the effects of consumer franchise-building (CFB) activities (primarily image advertising) versus non-CFB activities (such as price promotions) on market performance. He found that CFB activities had an impact on sales for about four years, while the effect of non-CFB tactics lasted for one year or less. Johnson (1984) looked at the relationship between advertising spending and brand loyalty. For those brands that suffered a decline in brand loyalty over time, one of the major contributing factors was a lack of advertising support.

It should be noted that in these studies, brand equity was not measured directly, but rather inferred, under the assumption that strong brand value translates into market share and profitability. But as Biel (1993) has observed, "While strong brands often have high market shares, market share alone does not distinguish them from other brands." Biel recommended that researchers focus more on the perceptual components of brand equity (especially brand image) and how they relate to consumer preferences.

For purposes of this project, we adapt the familiar hierarchy of effects model as a framework for studying various antecedents and consequences of brand equity from the perspective of the individual consumer. We suggest that consumers form perceptions about the physical and psychological features of a brand from various information sources. Their perceptions of the physical product could come from objective sources such as *Consumer Reports*, or from more subjective sources such as advertising or personal experience. Psychological distinctions, on the other hand, come primarily through advertising. (For many product/service categories, where actual differences between brands are minimal, psychological differentiation through advertising is the primary contributor to perceptions.) These perceptions, in turn, contribute to the meaning or value that the brand adds to the consumer—i.e., brand equity. Brand equity

then influences consumer preferences and purchase intentions, and ultimately brand choice. Figure 1 is a pictorial representation of these ideas and provides a guide for this investigation.

In summary, the importance of measuring and managing brand equity cannot be fully appreciated until we understand not only how equity is formed but also how it affects attitudes and behavior. Managers clearly need to be convinced of brand equity's impact on the bottom line. This research is a step in that direction.

Method

Two studies were conducted using essentially the same research design, but employing different product/service categories and different samples.

Selection of Categories and Brands

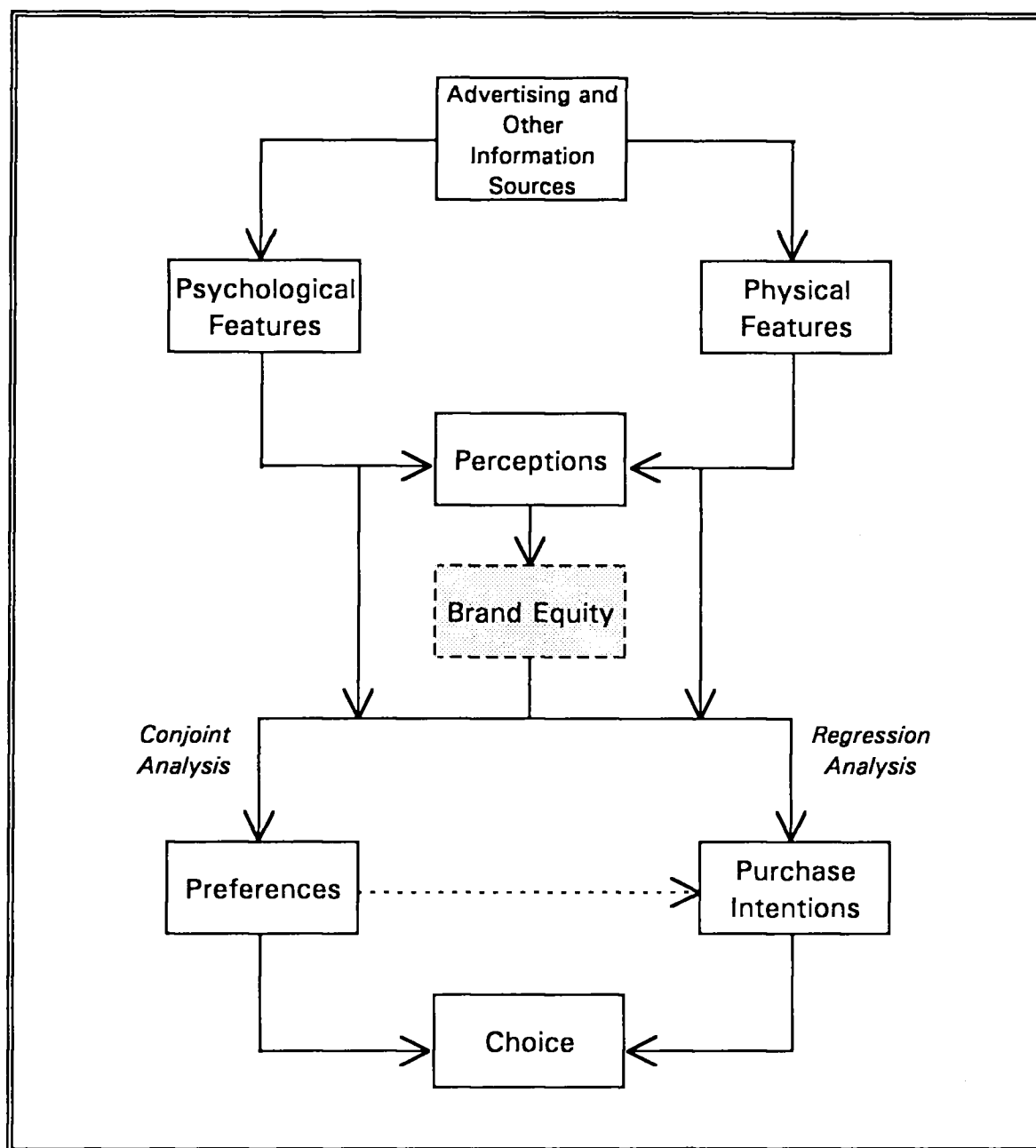
Three criteria were applied to the selection of categories. First, the study included both products and services, for comparative purposes. There are fundamental differences between goods and services which may have implications for brand equity. For example, services are less standardized and are composed largely of abstract, experience attributes, the values of which must be inferred by the consumer. Brand names, in this case, may be relied upon heavily for making such inferences. Unfortunately, very little empirical evidence exists on the equity of services (Smith 1991). One exception is the work of Landor Associates (1990), which listed two service organizations—Disney and NBC—among the top 10 most powerful brands in the U.S.

Second, the categories varied in functional and financial risk. Risk is one component of involvement which Landor Associates (1990) cited as a contributor to brand equity.

Third, the categories included pairs of brands that were highly similar on objective product attributes, but had dramatically different levels of advertising support. To make this determination, we consulted *Consumer Reports*, which rates brands across a wide range of industries, and *Leading National Advertisers* (LNA), which lists advertising expenditures by brand. Based on this secondary research, Holiday Inn and Howard Johnson were selected for the hotel category, and Soft Scrub and Bon Ami were selected for the household cleanser category.

Tables 1 and 2 present the respective brand ratings from *Consumer Reports*. As indicated, the pairs of brands yielded virtually identical scores on typical

Figure 1
Antecedents and Consequences of Brand Equity



evaluative criteria. However, the brand pairs were supported by dramatically different levels of advertising spending between 1980 and 1990. According to LNA data, Holiday Inn spent \$10.5 million in 1980; Howard Johnson spent \$600,000. In 1990, Holiday Inn's advertising budget was \$26.2 million, while Howard Johnson's budget was \$4.1 million. In the

cleanser category, Soft Scrub spent \$4.4 million on advertising during 1980; Bon Ami spent \$400,000. In 1990, the respective figures were \$7.4 million and \$100,000. Thus, during the 10-year period, Holiday Inn spent from 6 to 17 times more on advertising than did Howard Johnson. Soft Scrub spent between 11 and 74 times more than Bon Ami on advertising. It

Table 1
Consumer Reports Ratings of Hotels

Criterion	Brand	
	Holiday Inn	Howard Johnson
Overall Satisfaction Index	73 [*]	74
Cleanliness	3 ^{**}	3
Size of Room	3	3
Bed Comfort	3	3
Climate Control	3	3
Noise	3	3
Amenities (Linen & Toiletries)	3	3
Staff (Helpfulness & Efficiency)	3	3
Food Quality	2	2
Swimming Pool	3	3
Locations	East Midwest South West	East Midwest South
Typical Locales	Central City Highway Resort	Central City Highway Resort

^{*} Index of 100 indicates complete satisfaction. Differences of less than 4 points between brands are not meaningful.

^{**} On a scale of 1 to 5, where 1 = worse and 5 = better.

Source: Consumer Reports, September 1990.

was expected that these differences in ad spending would affect consumer perceptions of the brands and, ultimately, their brand equities.

Other factors, of course, could account for the differences in ad budgets. In the hotel category, for example, the different budgets could be attributable to the number of properties owned and operated by each

hotel chain. Holiday Inn has 1,365 properties, compared to 435 for Howard Johnson. Therefore, we adjusted the ad spending ratio according to number of properties. Holiday Inn still spent from 2 to 5 times more than Howard Johnson.

In the case of household cleansers, the different ad budgets could reflect differences in distribution and

Table 2
Consumer Reports Ratings of Cleansers

Criterion	Brand	
	Soft Scrub	Bon Ami
Overall Cleaning Ability	2*	2
Abrasiveness	Slight	Slight
Price	\$1.11	\$0.80
Size In Ounces	13 oz.	14 oz.
Cost Per Use	\$0.04	\$0.03
Stains Removed Easily	Tea Aluminum Copper Steel	Tea Aluminum Copper Steel
Type	Liquid	Liquid
Mars Surfaces If Left Unwiped?	No	No
Contains Bleach?	No	No
Strongly Alkaline?	No	No

* On a scale of 1 to 5, where 1 = worse and 5 = better.

** Source: Consumer Reports, January 1990.

product availability. Unfortunately, no national figures on cleanser distribution were available to the authors. Thus, we conducted a survey of 50 supermarkets in the surrounding metro area weighted by the stores' respective market shares. According to this informal survey, 90 percent of the local supermarkets carried Soft Scrub, while only 60 percent carried Bon Ami. Adjusting for product distribution, therefore, we found that the ad budget ratio was still 7 to 49 times greater for Soft Scrub than for Bon Ami.

Operationalization of Brand Equity

Brand equity can be measured in relative or abso-

lute terms. From the firm's perspective, absolute measures are probably more useful. Managers have an interest in maximizing their brand's equity. Thus, they need the ability to determine benchmarks and objectives for individual brands, which can then be contrasted against competitive offerings and industry norms.

However, when comparing a very limited number of brand equity scores (as in the present study), absolute measures are virtually meaningless. Furthermore, not all operationalizations allow the calculation of an absolute score. Therefore, this investigation relied on a relative measure.

The study employed the perceptual components of

Aaker's (1991) definition of brand equity, as advocated by Keller (1993). Recall that Aaker identified the following perceptual components: awareness (both brand and advertising), brand associations, and perceived quality. These dimensions were examined individually and then were combined to come up with an equity score for each brand. (In his book, Aaker [1991] recommends using a weighted average of the various components, but he does not say *how* to weight them. More than likely, the weighting scheme would vary from brand to brand. Given the lack of clarification, we chose to treat the components equally by using a simple average.)

Data Analysis

The first step in the data analysis was to verify the different levels of brand equity for the brands under investigation. The next step was to measure the effect of brand equity on consumer preferences, using the conjoint procedure. Conjoint analysis is a multivariate technique which determines the relative importance of a product's multidimensional attributes (Green and Wind 1975). We were not interested so much in the respondent's preference for predetermined attribute combinations (which is a common application of the conjoint technique) as we were interested in which brand yielded the higher preference/utility. Conjoint analysis also allowed us to measure the importance of brand name relative to other brand attributes. As Aaker (1991, p. 187) noted:

The name is the basic core indicator of the brand, the basis for both awareness and communication efforts. Often even more important is the fact that it can generate associations which serve to describe the brand—what it is and does. In other words, the name can actually form the essence of the brand concept.

To determine the attributes included in the conjoint analysis, a pretest was conducted among a separate sample of 28 users of cleansers and 45 users of hotels. In the pretest, subjects were asked to rank the attributes listed in *Consumer Reports* in order of determinancy, or importance as decision criteria, when choosing a brand. Because only a few attributes can be used in conjoint analysis (typically four to six), it was critical that they be both important and actionable (Churchill 1991). The top four attributes were included in the investigation, with brand name serving as the fifth attribute. Attribute levels were chosen to reflect industry practice. The scenarios were created using a fractional factorial design (Bretton-

Clark 1992).

The final step in the data analysis was to measure the impact of brand equity on brand usage intentions using multiple regression analysis. Again, we assumed that brand equity differences would be manifested in the brand name. The dependent variable in this analysis was intent to purchase the brand, measured on a 5-point scale. Respondents provided purchase intentions for the two brands in the respective product category. They also rated both brands on a number of evaluative criteria. For the hotel study, the independent variables were: brand name (dummy coded as 1=Holiday Inn, 0=Howard Johnson), price, room quality, cleanliness, staff quality, and food quality. For the cleanser study, the independent variables were: brand name (dummy coded as 1=Soft Scrub, 0=Bon Ami), price, abrasiveness, stain removal ability, effectiveness on various surfaces, and overall cleaning ability. All independent variables except brand name were measured on a 7-point scale.

A pure between-subjects design was used for the regression. Respondents in each study were split into two groups. For group one, we used only their evaluations of the first brand in the respective category. For group two, we used only their evaluations of the second brand. Because the subjects were in two groups, we were able to create a dummy variable that represented the brand each subject evaluated.

In summary, the data analysis utilized both a decompositional model (conjoint analysis), where respondents considered attributes simultaneously in the form of a profile, and a compositional model (regression analysis) in which respondents evaluated each attribute separately.

Sampling Procedure

The data for both studies were collected from users of the respective product or service category. Respondents were drawn on a voluntary basis from graduate and undergraduate business courses at a nontraditional commuter campus, where the average age of students is 27, and the vast majority work either full- or part-time. Students who expressed infrequent or no usage were asked not to participate in the project.

The hotel survey was administered to 90 evening MBA students, most of whom work full-time and, thus, are more likely to be familiar with hotels due to work-related travel. The hotel sample was 60 percent male, 65 percent single, and 89 percent white. The average age was 29. (According to data from *Simmons*

Market Research Bureau, or SMRB, the largest percentage of hotel users is 25-34).

The cleanser sample consisted of 92 users, broken down as follows: 59 percent were female, 72 percent were single, and 84 percent were white. The average age was 25. (According to SMRB, the largest percentage of users of scouring cleansers is 25-34.)

Research Instrument

The research instrument was divided into two parts, corresponding to the two purposes of the study: 1) measurement of brand equity, using the perceptual components of Aaker's (1991) definition; and 2) measurement of brand preferences and usage intentions.

Part I. The first question on the survey measured unaided awareness of brands in the respective product/service category. Respondents were asked to list as many brands as they could, from which were recorded: top-of-mind awareness of the brands under investigation and total brand mentions. This question was administered separately from the rest of the survey to ensure unbiased responses. Aided brand awareness and degree of brand familiarity were then measured.

The next set of questions elicited brand associations. Respondents were asked to list all descriptive words, thoughts, characteristics, symbols, or images that came to mind when the brand was mentioned. From this question, four variables were created: total associations, total positive associations, total neutral and total negative associations. Two trained researchers coded the responses. Using the procedure advocated by Perreault and Leigh (1989), interjudge reliability was calculated to be .90, with a lower confidence limit for reliability of .82.

Advertising awareness was measured next. Respondents were asked if they had ever seen any advertising for the respective brand. (It was decided not to put any time constraints on advertising exposure, since such a question would be difficult to answer and impossible to verify.) A follow-up question for subjects who reported advertising awareness asked for a description of what the advertising said or showed. Perceived quality was asked last in this section.

Part II. This section of the survey began with the conjoint questions. Respondents were asked to assume they were making a decision among brands in the respective category. They were then given 18 scenarios consisting of various combinations of five

attributes, with instructions to evaluate each combination on a 7-point semantic differential scale. The scenarios were presented in story form, with the attributes randomized within scenarios. To minimize a position bias, the order of presentation of scenarios was varied on three different forms of the survey.

The next set of questions served as input into the multiple regression analysis. Respondents were asked to rate five brands in the category (including the test brands) across various dimensions using a 7-point scale (from very bad to very good). In the hotel study, the dimensions were: price, room quality, cleanliness, staff quality, and food quality. In the cleanser study, the dimensions were: price, abrasiveness, stain removal ability, effectiveness on various surfaces, and overall cleaning ability. The instructions stressed that we were interested in perceptions and/or experiences (since perceptions about a brand can come from sources other than personal experience). Responses to these questions served the added purpose of helping us to verify that the sample's perceptions of brands matched the *Consumer Reports* ratings. In the final section, category and brand usage intentions were measured, along with a few simple demographic questions.

Results

Hotel Study

In the first study, we compared the equities of two hotel brands: Holiday Inn and Howard Johnson. As stated earlier, brand equity was measured using the perceptual components of Aaker's (1991) definition. The top portion of Table 3 presents the results. As indicated, Holiday Inn's brand equity was significantly greater than that of Howard Johnson. Holiday Inn was mentioned 8.5 times more than Howard Johnson in the unaided brand awareness task. (Interestingly, on an aided basis, both brands achieved almost perfect brand awareness scores. What this demonstrates is that Holiday Inn occupies the coveted location in the consumer's consideration set known as "top-of-mind." The Howard Johnson name is less readily accessible, but it is there in the consumer's memory, nonetheless.)

Eighty-seven percent of respondents stated that they had at some time seen advertising for Holiday Inn, compared to 56 percent for Howard Johnson. Thus, Holiday Inn had 55 percent more self-reported advertising awareness. The overall perceived quality of Holiday Inn was 1.74 times higher than that of Howard Johnson.

Holiday Inn had 4.75 times more positive associa-

Table 3
Hotel and Cleanser Brand Equity Using Aaker's Dimensions

Product Category	Brand	Unaided Brand Awareness	Advertising Awareness	Perceived Quality*	Brand Associations**			
					Total	Positive	Neutral	Negative
Hotels	Holiday Inn	85%	87%	5.4	4.0	1.9	1.8	0.3
	Howard Johnson	10%	56%	3.6	3.5	0.4	2.0	1.1
Cleansers	Soft Scrub	54%	62%	5.8	3.6	1.5	1.9	0.2
	Bon Ami	4%	3%	2.8	1.5	0.1	1.0	0.4

* On a scale of 1 to 7, where 1 = very bad and 7 = very good.

** Numbers represent averages.

tions than its competitor, while Howard Johnson had 3.66 times as many associations that were negative. By assigning a weight of +1 to positive associations, 0 to neutral ones, and -1 to negative associations, we determined that Holiday Inn generated a net of 8.41 more favorable thoughts, words, symbols, and images than did Howard Johnson, on average.

Looking at the specific brand associations made about the two hotels yields additional insight. Among the positive associations made about Holiday Inn were the following: *reputation* (updated image, good reputation, famous, distinctive, pervasive); *consistency* (uniform standards, no surprises); *service* (excellent, caring, hospitable, caters to customer needs); *value* (economical, good price); and *rooms* (clean, tasteful, comfortable, homey). Neutral associations about Holiday Inn included: *colors* (green, orange, yellow, white); *logo* (star, arrow, lights); many *price* mentions (mid-priced, moderate); *locations* (near cities, airports, interstates); and *specific features* (such as restaurant, bar, pool) presented in a non-evaluative manner. Holiday Inn also had some negative associations, such as: *lack of amenities* (bland, boring, no mints on pillow, homogenized); *noisy*; *rooms* (Sears-type decorating).

Howard Johnson's strengths, based on its positive associations, appeared to be: *value* (reasonable price, affordable); *good restaurant*; *good locations*; *convenient* (quick in and out); *safe*; and *comfortable*. Most of Howard Johnson's associations were neutral, including: *category* (motorlodge, roadside inn, chain, outside entrance); *colors* (famous orange and turquoise); *logo* (triangular roof, HoJo name); many *food* mentions (ice cream, clams, diner, waitresses in plaid, Shoney-type restaurant); *price* (moderate, mid-priced); and *locations*. Among the negative associations for Howard Johnson were: *reputation* (not spectacular, old, probably wouldn't use); *rooms* (minimum environment, gaudy colors, velvet, plastic, not renovated, not well maintained, boring, bland decor); *cleanliness* (stale, dark, dusty); and *restaurant* (truckstop, bland, greasy, cheap food).

Combining the various ratios (8.5 for unaided brand awareness, 1.55 for advertising awareness, 8.41 for net favorable associations, and 1.74 for perceived quality), we arrived at a brand equity score for Holiday Inn that was an average 5.05 times that of Howard Johnson. (It is worth noting that treating the components of brand equity in-

dividually or combining them into an overall brand equity score will not change the results, since the hotel which was higher overall in terms of equity was also higher on each individual dimension.)

Conjoint analysis was used next to examine the impact of brand equity on brand preferences. Table 4 presents the attributes included in the conjoint design, along with the attribute levels.

Results of the hotel conjoint analysis are shown in the top portion of Table 4. As indicated, brand name was fourth in feature importance, behind price, bed size and presence of a pool. Of greater interest for our purposes, the preference/utility for Holiday Inn (the brand with the higher equity) was 9.75 times greater than it was for Howard Johnson.

As a last step, regression analysis was used to investigate the association between brand equity and purchase intention. According to the results shown in Table 5, the coefficient for the dummy variable "hotel brand name" was 1.8. This variable was set to zero for Howard Johnson and set to one for Holiday Inn. Hence, the coefficient may be interpreted as the increment in purchase intention when choosing Holiday Inn over Howard Johnson. Stated alternatively, brand usage intention went up by 1.8 (on a 5-point scale) when switching from Howard Johnson to Holiday Inn. Clearly, the brand which had the higher equity also generated the higher purchase intention.

Overall, the results of Study 1 provide fairly convincing evidence of the effect of brand equity on brand preferences and usage intentions.

Cleanser Study

For comparative purposes, in Study 2 we examined a product category that is generally considered low in financial and functional risk—household cleansers. Again, the first step was to verify the different levels of brand equity for the two brands. The bottom portion of Table 3 presents the cleanser brand equity results using Aaker's definition. As the table reveals, Soft Scrub was mentioned 13.5 times more than Bon Ami in the unaided brand awareness task. (On an aided basis, brand awareness increased for both brands and the differential between the two was reduced somewhat.) Advertising awareness for Soft Scrub was 20.66 times more than that of Bon Ami. Overall perceived quality was 2.07 times greater for Soft Scrub than for Bon Ami.

Soft Scrub had 15 times more positive associations, while Bon Ami had twice the negative associations of Soft Scrub. Using the weighted procedure discussed

earlier, Soft Scrub generated a net of 17 more favorable associations, on average. Following are some of the characteristics, words, and symbols that were linked to Soft Scrub. Positive associations included: *reputation* (reliable, well established, safe); *ease of use* (convenient, quick, easy, neat, tidy); *effectiveness* (works great, hard-working, powerful, cleans well, tough enough to clean hard surfaces); *texture* (leaves no grit, soft, gentle, sensitive, creamy); *smell* (clean, fresh scent); *economical*.

Soft Scrub also generated many neutral, non-evaluative associations, particularly about product features, such as: *packaging*; *form*; *uses* (use in kitchen, bathroom, etc.); *category* (cleanser, disinfectant); *sudsing action*; and *texture*. Other neutral associations included: *color* (white, blue, bluish-green) and *effectiveness* (kills germs, removes stains). A few respondents mentioned *competition* (Dow with scrubbing bubbles). Negative associations about Soft Scrub centered around: *packaging* (hard to get out of bottle, messy top); *strong smell*; *texture* (gritty, grainy, watery); and *effectiveness* (leaves film residue, ruins clothes).

For the brand Bon Ami, positive associations (which were few) included: *friendly*; *reliable*; *secret ingredient behind success*; and *effectiveness* (good, hardworking, tough). Most of the associations tended to be neutral: *reputation* (around for a long time, older product, my grandmother used); *competition* (like Comet, like Ajax); *packaging* (shiny label, cylindrical container, generic can, white/gold/red on label, aluminum); *French*; *antiseptic*; *petite*; and *uses* (industrial, cleaning the house). Among the negative associations were: *reputation* (outdated, for old ladies, not reliable); *price* (cheap); *too abrasive*; *smell* (stinks); *texture* (gritty); and *effectiveness* (may not work well, messy to use, no fun).

Combining the various dimensions (13.5 for unaided brand awareness, 20.66 for advertising awareness, 17 for net favorable associations, and 2.07 for perceived quality), the brand equity score for Soft Scrub was an average of 13.31 times greater than that of Bon Ami. Again, it should be noted that the cleanser which scored higher overall in terms of brand equity also scored higher on each individual dimension.

Table 4 presents the conjoint design and results. From the table, it is evident that brand name had extremely high importance. In fact, this attribute was more important than any of the objective product features in determining preference for household cleansers. The preference/utility for Soft Scrub (the brand with the higher brand equity) was 33.7 times

Table 4
Conjoint Analysis for Hotels and Cleansers

Product Category	Feature	Feature Importance	Feature Levels	Utility of Levels
Hotels	Price	27.4%	\$35 \$55 \$75	13.2 12.5 1.0
	Bed Size	25.4%	Double Queen King	4.4 12.2 6.4
	Pool	19.4%	Yes No	9.3 0.2
	Times Cleaned Per Day	11.6%	Once More Than Once	0.7 5.6
	Brand Name	16.2%	Holiday Inn Howard Johnson	7.8 0.8
Cleansers	Price	18.7%	\$0.75 \$0.95 \$1.15	3.6 7.0 4.5
	Abrasiveness	21.4%	None Slight Moderate	4.7 7.8 8.0
	Number of Surfaces	12.6%	3 6	2.8 4.7
	Type	20.1%	Powder Liquid	0.5 7.5
	Brand Name	27.2%	Soft Scrub Bon Ami	10.1 0.3

greater than for Bon Ami.

Finally, we used regression analysis to estimate the effect of brand equity on brand usage intentions. The results in Table 5 reveal that the coefficient for brand name was 2.5, indicating that brand usage intention went up by 2.5 (on a 5-point scale) when switching from Bon Ami to Soft Scrub. Again, it is

clear that the brand with the higher equity generated the higher usage intentions.

Discussion

This study examined the effect of brand equity on consumer preferences and purchase intentions. For

Table 5
Regression Analysis for Hotels and Cleansers

Product Category	Independent Variable	Parameter Estimate	Significance Level	Overall R-Square
Hotels	Price	-1.10	.001	.50
	Cleanliness	0.30	.01	
	Room Quality	0.29	.02	
	Staff Quality	0.24	.03	
	Food Quality	0.20	.03	
	Brand Name	1.80	.0001	
Cleansers	Price	-0.30	.01	.45
	Cleaning Ability	0.30	.01	
	Stain Removal Ability	0.27	.02	
	Abrasiveness	0.26	.02	
	Effectiveness on Various Surfaces	0.20	.03	
	Brand Name	2.50	.0001	

comparative purposes, we included two sets of brands, one from a service category characterized by fairly high financial and functional risk, and one from a generally lower risk product category. Each set consisted of two brands that had invested very different levels of advertising spending over the past 10 years but were objectively similar in terms of physical attributes. (While every attempt was made to ensure that the brands were comparable, there were instances when this criterion could not be satisfied. For example, the two cleansers varied by type: liquid versus powder. This may have influenced results.)

As predicted, across both categories, the brand with the greater advertising budget yielded substantially higher levels of brand equity. In turn, the brand with the higher equity in each category generated signifi-

cantly greater preference and purchase intentions. Our findings confirm the conventional wisdom that advertising equals knowledge, and knowledge equals liking. While the various relationships have been shown by previous research in independent attempts, we think that this is the first study to demonstrate all of these relationships together in a brand equity framework. Moreover, we demonstrated these relationships in two very different product categories.

We cannot say definitively that advertising spending causes brand equity. Or that a lack of spending on advertising will destroy the value of a brand. However, if a marketer chooses to stop investing in the creation and maintenance of a brand franchise, then that marketer must be prepared for the prospect of losing equity over time. The problem is exacerbated if

competitors continue to invest in advertising. Advertising professionals are called upon daily to "prove" advertising's return on investment for the short term. This study offers some suggestion of advertising return on investment over the long term. While individual ad executions can be measured in terms of immediate cognitive, affective, and behavioral response, the real value of advertising can only be realized in the long-term. It should be noted that in the present study, we had access to only 10 years of advertising data. A more complete advertising history for each brand would no doubt yield even greater insights, since it takes years of advertising spending to create brand equity.

One of most interesting implications for advertising strategy is that products which are lower in risk and involvement *may* depend even more heavily on differences created through advertising than products at the other end of the spectrum. The difference in brand preference for high versus low equity brands was significantly greater for cleansers than for hotels.

It is also noteworthy that brand name had greater feature importance for cleansers than for hotels. Again, it could be that for high involvement products, consumers consider a wide range of features, with brand name being one of many attributes evaluated. For low involvement products where fewer features are likely to be evaluated, a brand name might serve as a "halo" through which consumers can make a quick assessment of the brand.

We expected brand name to have greater feature importance for services than for products, given the intangible, abstract nature of services and the fact that the name of a brand might serve as reassurance in the evaluation process. Our results did not bear this out. This finding should be viewed with caution, however, since the products and services used in the study were not at the same level of involvement.

Another notable finding concerns the relative importance of price in determining preferences. In the case of cleansers (where you might expect price to be determinant), brand name had significantly greater feature importance than did price. This would suggest caution in the rush to shift spending from advertising to promotion. As the saying goes, "Promotion builds volume; advertising builds value." And while price cuts can be met by the competition, the meaning of a brand created through advertising is less susceptible to encroachment by competitors.

Looking more closely at the relationship between advertising spending and brand associations, a sig-

nificant finding emerged. We would expect an increase in advertising to contribute to an increase in total associations. This was true with cleansers. But for hotels, Howard Johnson had almost as many total associations as did Holiday Inn. It was more the *mix* of associations that varied so much in the hotel category. Most of Howard Johnson's associations were neutral or negative. Certainly, one of the strengths of advertising is that it allows the sponsor to control the message, and thus create positive associations. The question is worth considering: If Howard Johnson increased its ad budget, would its positive associations go up?

A related issue concerns the content of advertising. In this study, we looked at advertising in terms of sheer dollar volume. But undoubtedly, the nature and quality of advertising content play a role in the perceptual components of brand equity and may influence behavioral manifestations.

Other variables besides advertising could, of course, account for the differences in brand equity scores. This is a limitation of the study. While we tried to control for some of the variables (such as distribution), there was no way to control every conceivable dimension, especially in a field investigation. For example, we did not take into account the use of sales promotion tools such as couponing and price discounts. As another example, it could be argued that age differences in the cleanser brands might account for the findings. (Holiday Inn and Howard Johnson were founded at about the same time—1952 and 1954, respectively. So this was not a variable in the hotel category.) In other words, a demographically older sample might have generated stronger associations for the older brand, Bon Ami. However, one could just as easily argue that an older brand has had more time to establish brand equity and, thus, should have an advantage over a newer brand.

An additional limitation of this project concerns the use of a convenience sample made up of students. This was not viewed as an overwhelming problem since the students were users of the respective categories. However, it could be argued that the overall sample in each case was not the primary target for the product category or the individual brands. We do not know for certain who the targets are for the respective brands, but we can determine the profiles of current brand and category users. Using a combination of SMRB and MRI data, we were able to confirm that our sample fits fairly closely with the profile of category users. In addition, we compared the user profiles for each pair of brands. The differences were

nominal. Howard Johnson users were *slightly* more likely to be black, older, live in the mid-Atlantic, and have less college education. Soft Scrub users were *slightly* more likely to be single, employed, have higher income and live in the suburbs.

In conclusion, the measurement and management of brand equity have become top priority marketing issues in recent years, as evidenced by the growing literature on the subject. Most articles automatically assume that brand equity has an impact on a brand's performance, ergo brands should do everything feasible to increase their equity. However, it does not make sense economically to invest a firm's scarce resources on strategies to add value if the value doesn't translate into preferences and purchase behavior. Firms need empirical evidence of the consequences of brand equity. The present study demonstrated that brand equity increases both consumer preferences and purchase intentions. Future studies should examine more closely the antecedents of brand equity, particularly the role that advertising plays in adding value to the brand and in helping "great brands live forever."

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